Arunabha Majumder

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Portfolio: arunabha22.github.io/arunabha-majumder

CYBEX, Bayreuth

Footwear Innovation Engineer (m/f/d)

Dear CYBEX team,

I'd like to apply for the Footwear Innovation Engineer role. I have spent the last 6 years building things that people wear, from an exo glove to a shoulder exoskeleton, and testing them on real bodies. Making children's shoes that fit and support growing feet better is the kind of work where that experience would be useful.

As a research assistant at Aalborg University I designed a hybrid shoulder exoskeleton, with a spring mechanism and a 6 Nm motor working together to lift the arm. I optimised it with a genetic algorithm, designed it in SolidWorks, 3D printed prototypes, had the final parts manufactured and measured its stiffness on a physical test setup. It uses over 25% less motor energy and won the Best Paper Award at ISRM 2026. In a study with this exoskeleton I collected EMG data from 12 participants using the Qualisys system.

I also designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM), estimated its stiffness with less than 10% error and quantified human arm impedance in a study with 6 participants, where Qualisys cameras recorded the arm position data. So setting measurable targets, testing against them and refining the prototype is how I normally work.

In my Master's I fabricated silicone-based pneumatic muscles for a soft exosuit in a team of 4, and in my Bachelor's I built an arm exoskeleton and an exo glove for a paralysed arm. I also hold a Senior Diploma in Fine Arts and do graphic design (Illustrator, Photoshop, Procreate), so I sketch ideas by hand and like working closely with designers.

I'm ready to move to Germany and I'm willing to learn German. I would love to bring new and innovative ideas to your footwear and grow with the team. My portfolio, with videos and results from my work, is at arunabha22.github.io/arunabha-majumder. Thank you for reading.

Best,
Arunabha Majumder